

Characterisation of Aggregated Electric Thermostatic Loads and Quantification of their Flexibility

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Abstract—Distribution system operators (DSOs) rarely know how many electric thermostatic loads (ETLs), such as heat pumps (HPs), are connected to their low-voltage networks. This paper estimates the installed capacity, temperature-driven energy and flexibility potential of an aggregated ETL population from the net load and ambient temperature a DSO already holds. A piecewise-linear fit of the net load against temperature gives the temperature-driven ETL demand. A simultaneity factor (SF) fit, calibrated on a submetered pilot, converts that demand into the installed capacity, as the ratio of the two temperature sensitivities. Only the SF sensitivity transfers from the pilot, while each target keeps the threshold temperature of its own net-load fit. On 1260 synthetic aggregates of Swiss households, the installed capacity is recovered at a weighted absolute percentage error (WAPE) of 19%, falling to 12–14% for aggregates of fifty consumers. On three real Swiss and German aggregates outside the pilot, the estimates lie within 10% of the measured capacity. The temperature-driven energy needs no pilot and is recovered at a WAPE of 9%, and the flexible energy at 10% against an hourly submetered reference. Across aggregates from four countries, the SF sensitivity groups by climate class, so the pilot should share the target's climate and ETL technology.

Index Terms—Electric thermostatic loads, heat pumps, low-voltage networks, simultaneity factor, installed capacity, smart meter data, flexibility, distribution system observability.

NOMENCLATURE

ETL	Electric thermostatic load.
P_{net}	Aggregated net load.
P_{ETL}	Aggregated ETL load.
$P_{\text{non-ETL}}$	Aggregated non-ETL load.
P_{base}	Temperature-independent base load.
$\hat{P}_{\text{ETL}}^i$	Fitted ETL load of aggregate i .
s_h, s_c	Heating, cooling sensitivities of the net-load fit.
T_h, T_c	Heating, cooling threshold temperatures.
$P_{\text{ETL}}^{\text{max}}$	ETL installed capacity.
P_k^{max}	ETL installed capacity active in regime k .
SF_h, SF_c	Heating, cooling simultaneity factor.
$\hat{\text{SF}}_h, \hat{\text{SF}}_c$	Fitted heating, cooling simultaneity factor.
b_h, b_c	SF intercepts at the threshold temperatures.
m_h, m_c	Heating, cooling SF sensitivities.
R_h^2, R_c^2	Coefficient of determination of a fit on the days of the heating, cooling regime.
α_F	Flexibility duty cycle.
Δt	Energy integration period.
Δt_{Th}	Thermal-inertia window.
E_{ETL}	Temperature-driven ETL energy.
E_{flex}	Flexible energy.
N, N_{hp}	Number of consumers and heat pumps.

I. INTRODUCTION

The electrification of residential heating and cooling is a central lever of decarbonisation policy. Its instruments are electric thermostatic loads (ETLs), such as heat pumps (HPs), electric resistance heaters and air conditioners, and the HP leads their uptake [1]. With them grows the share of household demand that follows outdoor temperature. On 200 German LV feeders, for example, installed HP power rose by over twenty per cent from 2023 to 2025 [2].

This uptake supports decarbonisation, yet it loads low-voltage (LV) networks that were not sized for it. Field and simulation studies show that ETLs raise coincident feeder and transformer peaks in extreme weather. For HPs, the most studied case, these peaks are sharpest in cold weather and add to those of electric vehicles and rooftop solar [3]–[7]. To plan and operate these networks, network operators must know how much thermostatic load sits behind each metering point.

A distribution system operator (DSO) rarely holds that information. It meters aggregated net load and can obtain weather data. It seldom holds the consumer count, appliance registers or nameplate ratings needed to interpret the thermal response [2], [8]. The installed capacity of the ETLs behind a feeder therefore stays largely unobserved.

Four lines of research approach this gap, summarised in Table I. Non-intrusive load monitoring and HP detection recover thermostatic load from customer meters, either as a present-absent label [9], [10], [29] or as a disaggregated energy time series [11], [12]. Some extend to LV substations [14], [15]. These methods, however, need labelled or submetered training data, stop short of the installed capacity, and remain largely synthetic at the substation.

Change-point and degree-day models fit load as a piecewise-linear function of temperature. They range from canonical building-level formulations [30]–[32], reviewed in [33], to aggregated and smart-meter populations [16]–[18]. They recover heating and cooling sensitivities and a threshold temperature without any labels. They stop, however, at the coincident response and never reach the non-coincident installed sum.

For photovoltaics (PV), by contrast, installed-capacity estimation is well developed. Methods infer the capacity from aggregated net load or smart-meter data [22], [23], [34], [35], and even recover the service-transformer rating this way [24]. Irradiance plays there the role that temperature could play for ETLs, yet the thermostatic-load counterpart is largely missing [8].

The simultaneity of an aggregated ETL population, and the coincident peak it produces, inform network planning, as studied mostly for HPs. They set the design load [25], [26], [36], [37] and bound the flexibility the population offers [28], [38]. Planners measure these quantities on submetered populations of known size and read them as a planning input, rather than recover them from an un-submetered feeder.

To the best of the authors' knowledge, no prior method estimates the installed capacity of an aggregated ETL population from feeder-level net load without submetering the target feeder. The change-point models observe the right signal but lack the step from coincident demand to installed capacity, while the simultaneity studies hold that step but only on submetered populations. This paper joins the two. The contributions are threefold.

- 1) A net-load fit, piecewise linear in ambient temperature, characterises the ETLs behind a metering point from its aggregated net load alone. Its arms give the ETL temperature sensitivities, its thresholds the temperatures at which the population starts heating and cooling, and its flat centre the temperature-independent load. Integrating the heating and cooling arms estimates the temperature-driven ETL energy without any submetering.

TABLE I
PRIOR WORK ON DETECTING AND QUANTIFYING DERs

Reference	DER	Task	Level	Method	Input	Installed capacity	Submetering needed	Transferable
<i>Detection and disaggregation</i>								
[9]	HP	Detection	Meter	ML (PU learning)	Smart meter		✓	
[10]	HP	Detection	Meter	ML	Smart meter + weather		✓	
[11]	HP	Disaggregation	Meter	ML (deep)	15-min smart meter		✓	
[12]	HP	Detection + disaggregation	Meter	ML (CNN)	Smart meter		✓	
[13]	PV/EV/HP	Detection	Meter	ML	Net demand		✓	
[14]	Loads/DER	Regional NILM	Substation	NILM	Substation load		✓	
[15]	Loads	Disaggregation	Substation	ML (deep)	Substation load		✓	
<i>Temperature-response models</i>								
[16]	Cooling	Change-point model	Community	Change-point	Aggr. load + temp.			
[17]	HVAC	Change-point model	Meter	Change-point	Smart meter + temp.			
[18]	Sector load	Weather effects	System	Regression	Sector load + weather			
<i>PV and transformer capacity</i>								
[19]	PV	Detection + capacity	Meter	ML	Smart meter	✓	✓	
[20]	PV	Capacity + energy	Meter	ML	Smart meter	✓	✓	
[21]	PV	Disaggregation	Feeder	Physics	Power flow + GHI			
[22]	PV	Capacity	Feeder	Probabilistic	Aggr. net load	✓		
[23]	PV	Capacity	Substation	ML / model-based	Aggr. net load + GHI	✓		✓
[24]	Transformer	Nameplate rating	Transformer	Model-free	Smart meter	✓		
<i>HP coincidence, profiles and flexibility</i>								
[6]	PV/EV/HP	Coincidence / impact	Feeder	Empirical	Smart-meter profiles			
[25]	HP	Coincidence / ADMD	Population	Field study	Submetered HP		✓	
[26]	HP	Peak in cold weather	Population	Field study	Submetered HP		✓	
[27]	HP	Load profiles	Feeder	Physics (thermal)	Temp. + housing			✓
[28]	HP	Flexibility / DR	Population	Field study	Submetered HP		✓	
This work	HP / ETL	Capacity, energy, flexibility	Feeder	Change-point + SF	Net load + temp. + pilot SF	✓		✓

Installed capacity: recovers the installed capacity. Submetering needed: needs submetered device data or labels. Transferable: a calibration transfers to un-submetered aggregates. Level: individual meter, feeder, substation or transformer, or a submetered population.

- 2) The installed ETL capacity is estimated per regime as the ratio of the net-load sensitivity to the sensitivity of a simultaneity factor (SF) fit. Only this SF sensitivity is transferred from a submetered pilot of the same population, while each target keeps the threshold of its own net-load fit. The transfer is tested on synthetic Swiss aggregates drawn without replacement and on real Swiss and German aggregates outside the pilot.
- 3) The flexibility potential of the aggregated ETLs is defined as the energy they can shift in time within a thermal-inertia window, while still delivering the same total energy. It reduces to the fitted ETL load times the headroom left by the transferred SF, so it too needs no submetering of the target, and it is validated against an hourly submetered reference.

The rest of the paper is structured as follows. Section II states the problem, defines the net-load and SF fits and characterises them across datasets; Section III derives the energy, installed capacity and flexibility; Section IV evaluates them on synthetic aggregates of one Swiss population; Section V applies the method to real Swiss and German aggregates outside the pilot; Section VI discusses accuracy, generalisation and data requirements. Lastly, Section VII concludes the paper.

II. PROBLEM AND APPROACH

Privacy concerns and limits on metering and data exchange leave DSOs without full observability of individual consumers. Their records of distributed energy resources (DERs) also lag behind what is actually connected. DERs that consume or generate with a high simultaneity factor matter most for reliable operation, since they add to the peak together. This work therefore uses the load profiles a DSO does hold, aggregated in space at the feeder head or substation and

in time over the day. From these, it estimates the installed capacity and flexibility potential of the thermostatic loads behind the meter.

A metered point of an LV network, such as a substation or feeder, aggregates the load of the consumers downstream. This paper calls such a metered point, together with the consumers behind it, an aggregate. Some of its consumers run ETLs. The task is to quantify those ETLs from the aggregate's net load.

The net load is the sum of an ETL and a non-ETL component,

$$P_{\text{net}}(t) = P_{\text{ETL}}(t) + P_{\text{non-ETL}}(t), \quad (1)$$

where only the ETL component responds directly to temperature. The non-ETL component also carries a weaker, indirect temperature dependence, visible in the whole-home energy signature [39]. Still, ETLs dominate the total temperature response, so the model neglects the non-ETL contribution. Section VI revisits this assumption.

The problem is to recover the aggregated ETL load $P_{\text{ETL}}(t)$ from the aggregated net load. From it follow three quantities that aid DSO operation and planning: the ETLs' installed capacity, their energy consumption and their flexibility potential. Fig. 1 sketches the pipeline.

Two fits describe the ETL population's response to ambient temperature. The first, the net-load fit, uses data the DSO already holds, with no submetering. The second, the SF fit, needs submetered ETL data. Both share the same piecewise-linear form and use daily means, which reduce the effect of building thermal inertia [17].

A. Net-Load Fit

When an aggregate's net load is averaged over each day and plotted against the daily mean temperature, it follows a piecewise-linear pattern. The load is roughly flat on mild days, and rises close to

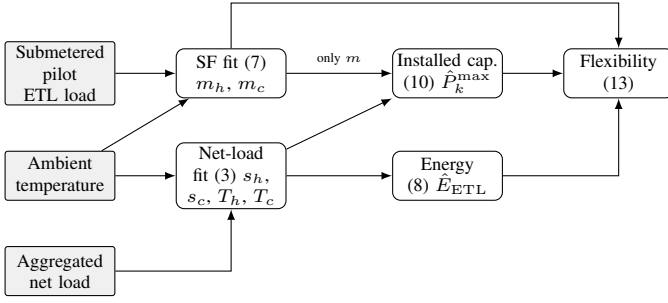


Fig. 1. Method pipeline. The observable net-load fit supplies the sensitivities s_h , s_c and the thresholds T_h , T_c ; the SF fit of a submetered pilot supplies only the SF sensitivities m_h , m_c . Their ratio gives the installed capacity per regime. The net-load fit alone gives the temperature-driven energy, and the capacity with the transferred sensitivity gives the flexibility.

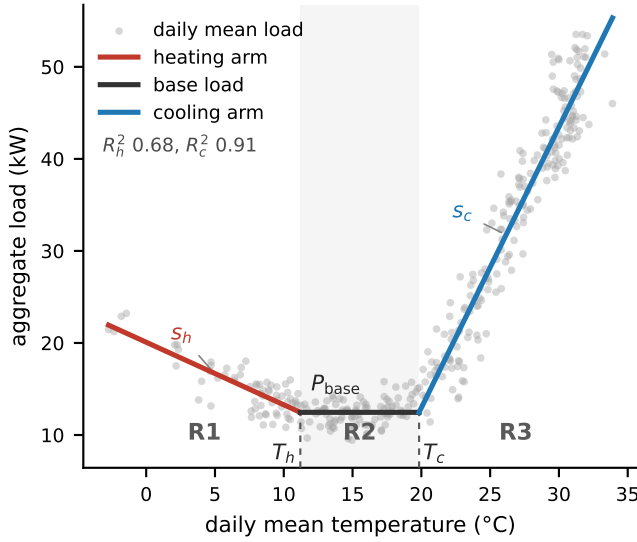


Fig. 2. Net-load fit (2) to the daily mean load of an aggregate of air-conditioned and furnace-heated dwellings in Austin, Texas [40], with the heating (R1), dead-band (R2) and cooling (R3) regimes.

linearly as the temperature falls below, or rises above, a threshold. The net-load fit expresses the aggregated net load of (1) as

$$\hat{P}_{net}(T) = P_{base} + s_h \max(0, T_h - T) + s_c \max(0, T - T_c), \quad (2)$$

where P_{base} is the temperature-independent load [kW], s_h and s_c are the heating and cooling thermal sensitivities [kW/°C], and $T_h \leq T_c$ are the heating and cooling threshold temperatures [°C] bounding a dead band over which no thermal response is present. The parameters of the fitted curve are computed through Ordinary Least Squares (OLS), with T_h and T_c being the values that minimise the squared residuals of the entire fit. The two temperature-dependent arms are the fitted aggregated ETL load.

Note that (2) is an observable fit: it needs only the net load and ambient temperature the DSO already holds, and no submetered or labelled ground truth. Fig. 2 shows the net-load fit to aggregated Austin smart-meter data [40], with the parameters of (2) marked. The thresholds divide the temperature axis into three regimes. R1 (below T_h) and R3 (above T_c) are the heating and cooling regimes, and R2 is the dead band between them, where the population neither heats nor cools. This work characterises R1 and R3 and treats R2 as out of scope.

For an aggregate i with its own fitted parameters,

$$\hat{P}_{ETL}^i(T) = s_h^i \max(0, T_h^i - T) + s_c^i \max(0, T - T_c^i), \quad (3)$$

where the hat marks a fitted function rather than a measurement. This estimated load only considers the temperature-driven component of ETL. The flat term P_{base} is the temperature-independent load, dominated by the non-ETL component of (1), and is discarded.

The fitted parameters carry a physical interpretation. The threshold temperatures T_h and T_c are behavioural. They mark where the population, on average, begins to heat or cool, rather than a fixed property of any dwelling. Below T_h a heating load appears and above T_c a cooling load appears, each growing close to linearly with the distance from its threshold. Between the two lies a dead band where the population neither heats nor cools. Individual dwellings start heating or cooling at different temperatures. Once many are aggregated, these differences average out, and T_h and T_c settle at values that reflect regional preferences for starting space heating or cooling (Section II-C).

The sensitivities s_h and s_c of the net-load fit measure how much heating or cooling load the population adds per degree. They scale with the ETL capacity installed behind the meter, so an aggregate with more ETLs shows a steeper arm, all else equal. Between aggregates of the same size, a steeper arm therefore also indicates a higher ETL rate, i.e., a larger share of consumers with an ETL. Section IV-B confirms this, with s_h tracking installed capacity across the aggregates.

The flat centre P_{base} is the

load that remains on days between T_h and T_c . The net load on these days is not constant, but it shows no trend with temperature. This temperature-independent load is outside the scope of this paper.

The balance between s_h and s_c reflects the local climate. Hot regions are expected to be cooling-dominated, with $s_h \approx 0$, whereas regions with long winters, such as the Nordic countries, are heating-dominated, with $s_c \approx 0$. Table II spans this range, from the heating-only Hamelin and Kloten aggregates to Maricopa, where cooling dominates.

B. Simultaneity-Factor Fit

The simultaneity factor (SF), also called the coincidence factor, is the ratio of the demand of a group of loads, here ETLs, at a given instant to the sum of their individual maximum demands. It reflects that connected devices do not all draw their full power at the same instant [41]. In distribution networks, the SF is central to planning. It lets utilities size transformers, cables and protection for the peak the group actually draws, rather than for the sum of individual peaks.

Mathematically, the SF is the fraction of the aggregate's installed capacity consumed at a given temperature T , and is given as

$$SF(T) = \frac{P_{ETL}(T)}{P_{ETL}^{max}}, \quad (4)$$

where $P_{ETL}(T)$ is the daily mean ETL load on a day of mean temperature T , and P_{ETL}^{max} is the cumulative installed capacity of all the ETLs in the aggregate. In this work, the installed capacity of each ETL is taken as the 99.9th percentile of its submetered 15-min draw. This tracks the rated power while ignoring isolated metering spikes.

An aggregate may have a mix of ETLs with different behaviours with respect to temperature. While some ETLs can be active in different regimes of the net-load fit, such as reversible HPs that can heat and cool, others are exclusive to either heating (e.g. resistive heating) or cooling (e.g. air-conditioning). Using the total ETL installed capacity in (4) would lead to an underestimation of the SF, since the installed capacity of cooling-exclusive devices would still

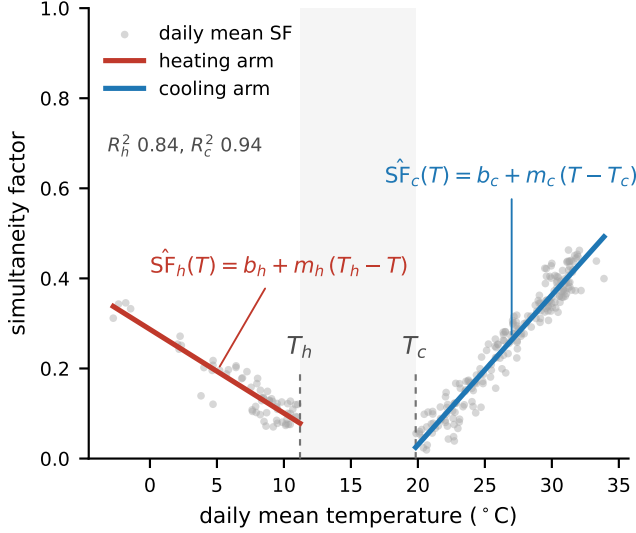


Fig. 3. SF fit (7) to the aggregated Austin ETLs [40], with each daily point normalised by the installed capacity of the ETLs active in its regime (6).

count towards the SF in the heating regime, and *vice-versa*. Therefore, each regime will have its own capacity, which is defined as

$$P_k^{\max} = \sum_{n \in \mathcal{N}_k} P_n^{\max}, \quad (5)$$

where k denotes the regime (heating or cooling), $\mathcal{N}_k$ is the set of ETLs active in regime k , and $P_n^{\max}$ is the installed capacity of ETL n . The true value of SF is then defined per heating and cooling regime as

$$\begin{aligned} \text{SF}_h(T) &= P_{\text{ETL}}(T) / P_h^{\max}, \quad \forall T < T_h, \\ \text{SF}_c(T) &= P_{\text{ETL}}(T) / P_c^{\max}, \quad \forall T > T_c, \end{aligned} \quad (6)$$

with h denoting heating and c cooling. It is by definition bounded in $[0, 1]$. The SF fit takes the same form as (2),

$$\begin{aligned} \hat{\text{SF}}_h(T) &= b_h + m_h(T_h - T), \quad \forall T < T_h, \\ \hat{\text{SF}}_c(T) &= b_c + m_c(T - T_c), \quad \forall T > T_c, \end{aligned} \quad (7)$$

where T_h and T_c are the thresholds of the net-load fit (2) of the same aggregate. The SF fit does not fit thresholds of its own. It borrows them from the net-load fit, so both fits describe the same heating and cooling regimes. The intercepts b_h and b_c and the heating and cooling SF sensitivities m_h and m_c [$1/^\circ\text{C}$] are determined by OLS on the days of each regime. The intercepts capture the weather-independent share of ETL use, such as domestic hot water, which the net-load fit instead places in P_{base} . Fig. 3 shows the SF data points and the SF fit for the same Austin population of Fig. 2.

The observed SF stays well below 1. Each arm would reach the installed capacity of its regime, with every ETL drawing at once, only at temperatures well outside any observed weather. The gap between the observed ETL load and the installed capacity is the headroom that the flexibility of Section III-C draws on.

C. Cross-dataset characterisation

Table II reports both fits, $\hat{P}_{\text{net}}(T)$ and $\hat{\text{SF}}(T)$, on aggregates from four countries and a range of climates, with each climate given as its Köppen–Geiger class [48]. R1, R3, or both are resolved as the local climate dictates, and each is scored on the days of its own regime. R1 fits with a net-load R_h^2 between 0.68 and 0.92 across the year-round

aggregates, lowest in Austin where electric heating is marginal. R3 ranges from R_c^2 of 0.95 in hot Maricopa to 0.75 in mild Oregon, where cooling is rare.

The threshold temperatures follow the local climate. T_h falls between 8 and 17 $^\circ\text{C}$, and T_c between 18 and 26 $^\circ\text{C}$. A hot county such as Maricopa carries a steep R3, whereas a cold one such as Hamelin carries none. The sensitivities s_h and s_c of the net-load fit scale with the aggregate's total installed ETL, and therefore with its size. The 849-dwelling Norwegian pool and the electrified ResStock counties show the steepest arms for this reason. The SF sensitivities m_h and m_c remove this size dependence.

The SF sensitivities m_h and m_c group partly by Köppen class. The two temperate oceanic (Cfb) aggregates, Hamelin and the Klotten pool, sit at $m_h = 0.016$ and $0.018 \text{ } ^\circ\text{C}^{-1}$, despite running on water-source and mixed ground-and-air-source heat pumps respectively. The two humid continental (Dfa and Dfb) sites, Oslo and Hennepin, converge on $m_h \approx 0.022 \text{ } ^\circ\text{C}^{-1}$, even though Oslo's heating load is mostly ground-source and direct resistance, while Hennepin's is air-source. The temperate dry and warm summer (Csb) class, however, splits. Its two measured aggregates, the Washington and Oregon homes with ductless HPs, sit at $m_h = 0.023\text{--}0.024 \text{ } ^\circ\text{C}^{-1}$, whereas the simulated King County aggregate of ducted HPs with resistance backup reaches $0.036 \text{ } ^\circ\text{C}^{-1}$. The cooling sensitivity m_c spans $0.024\text{--}0.038 \text{ } ^\circ\text{C}^{-1}$ without a clear grouping. The hot desert site, Maricopa, is the outlier, with the steepest heating sensitivity of all, $m_h = 0.063 \text{ } ^\circ\text{C}^{-1}$. This is possibly because a single aggregate and few genuinely cold days leave little data to anchor the coldest end of its heating arm. Overall, the Köppen class narrows m_h for aggregates of similar ETL technology, but the technology, and possibly the difference between measured and simulated peaks, matters as well. Across classes, m_h varies by up to a factor of four, so the transfer of Section II-D is only expected to hold within a climate class and ETL technology.

D. Transferability of SF

The SF is computed from submetered ETL data, which a DSO rarely holds for the aggregate under study. The SF fit (7) is therefore calibrated on a submetered pilot and reused on aggregates of the same population that have no submetering. Only the SF sensitivities m_h and m_c transfer. The pilot's thresholds come from the pilot's own net-load fit and follow its building stock and climate, so the target's thresholds T_h and T_c are read from the target's net-load fit instead. The pilot's intercepts b_h and b_c describe weather-independent ETL use, which the target's net-load fit places in P_{base} and never attributes to its ETL arms.

The transfer rests on one hypothesis: within a population, m_h and m_c do not track the installed capacity or the ETL rate of the aggregate. Section IV-C tests this hypothesis within the Klotten population, and Section V tests it across populations. If it holds, the transferred sensitivity turns the observable net-load fit into an estimate of the unobserved installed capacity, through (10). This is the mechanism the rest of the paper builds on.

III. AGGREGATED ETL QUANTIFICATION

The two fits, (3) and (7), are complementary. The net-load fit (3) is observable, but its sensitivities also absorb any temperature response of the non-ETL load. The SF fit (7) needs submetering but isolates the ETL simultaneity. Together they estimate three quantities a DSO needs about the ETLs of an aggregate: their energy consumption, installed capacity and flexibility potential. Energy consumption follows from (3) alone. Installed capacity and flexibility also need the transferred SF sensitivity, since the net load exposes the aggregated

TABLE II
NET-LOAD AND SF FITS, ONE AGGREGATE PER DATASET

dataset	ETL tech	n	Climate	Net-load fit $\hat{P}_{\text{net}}(T)$							SF fit $\hat{S}\hat{F}(T)$							
				P_{base}	s_h	T_h	R_h^2	s_c	T_c	R_c^2	b_h	m_h	SF_h^{max}	R_h^2	b_c	m_c	SF_c^{max}	R_c^2
Hamelin, DE [42]	WSHP	37	Cfb	16	3.9	13.9	0.72	–	–	–	0.014	0.016	0.32	0.72	–	–	–	–
Kloten, CH [43], [44]	ASHP, GSHP	50	Cfb	54	6.1	16.5	0.92	–	–	–	0.029	0.018	0.47	0.95	–	–	–	–
Oslo, NO [45]	GSHP, ER	849	Dfb	516	52.5	15.2	0.88	–	–	–	0.106	0.022	0.70	0.80	–	–	–	–
Hennepin, MN, US [46] ^a	ASHP	150	Dfa	118	42.3	8.0	0.74	–	–	–	0.061	0.022	0.46	0.93	–	–	–	–
WA, US [47] ^b	DHP, ER	22	Csb	25	2.4	14.6	0.90	2.0	21.3	0.79	0.058	0.024	0.53	0.91	0.067	0.031	0.24	0.83
OR, US [47] ^b	DHP, ER	21	Csb	25	2.1	13.3	0.87	2.0	21.8	0.75	0.044	0.023	0.46	0.91	0.097	0.024	0.22	0.79
King, WA, US [46]	ASHP, ER	150	Csb	146	21.1	13.3	0.89	11.0	18.4	0.93	0.036	0.036	0.53	0.90	0.091	0.038	0.37	0.96
Austin, TX, US [40]	AC, ER	22	Cfa	12	0.7	11.2	0.68	3.0	19.8	0.91	0.078	0.019	0.34	0.84	0.025	0.033	0.49	0.94
Maricopa, AZ, US [46]	ASHP, ER	150	BWh	172	15.3	11.6	0.83	25.7	26.0	0.95	0.144	0.063	0.52	0.86	0.047	0.024	0.39	0.95

n: aggregated consumers. Climate: Köppen–Geiger class [48]. R_h^2 , R_c^2 : fit scores in R1 and R3. ASHP, GSHP, WSHP: air-, ground-, water-source HP; DHP: ductless (mini-split) HP; AC: air conditioning; ER: electric resistance heating.

^a Hennepin's SF uses the ASHP compressor's own draw only, excluding its electric-resistance HP-backup element

^b The SF uses the submetered ductless HPs only. Most of these homes also have electric baseboard, furnace or other zonal heating circuits (17 of 22 in WA, 16 of 21 in OR), which the net load includes.

ETL demand, not the non-coincident sum of the individual peaks. Section IV evaluates each quantity.

A. ETL Energy Consumption Estimation

The energy the ETL population draws follows from integrating its temperature response over a certain period. Over a day of mean temperature T , the temperature-driven ETL energy is given by

$$\hat{E}_{\text{ETL}} = (s_h \max(0, T_h - T) + s_c \max(0, T - T_c)) \Delta t, \quad (8)$$

with Δt being the length of the period, here one day. The annual energy is the sum of (8) over the days of the year. The fitted ETL load alone suffices. Given a measured temperature, it recovers the aggregated ETL load $\hat{P}_{\text{ETL}}(T)$ from the net load $P_{\text{net}}(t)$; given a forecast temperature, it forecasts that load for a future period.

R2 bounds the estimate. Between T_h and T_c both arms are zero. The estimate therefore misses any ETL energy that persists into mild weather, such as domestic hot water. The estimate captures the temperature-driven share, not the annual total.

B. ETL Installed Capacity Estimation

The installed capacity is the sum of the individual peak demands (5), defined per regime. The aggregated ETLs never draw it at once, so the net load never exposes it directly. The net-load fit does expose the temperature-driven ETL load $\hat{P}_{\text{ETL}}^i(T)$ of aggregate i . Rearranging (6), the capacity of regime k follows from this load and the SF at the same temperature.

Both must describe the same load. The net-load arm is zero at the aggregate's own threshold by construction, and the weather-independent ETL use sits in P_{base} . The matching SF is therefore the temperature-driven part of (7). It starts from zero at the aggregate's own thresholds and grows with the transferred sensitivity,

$$\begin{aligned} \hat{S}\hat{F}_h^i(T) &= m_h (T_h^i - T), \quad \forall T < T_h^i, \\ \hat{S}\hat{F}_c^i(T) &= m_c (T - T_c^i), \quad \forall T > T_c^i. \end{aligned} \quad (9)$$

Dividing the arms of (3) by (9) gives the installed capacity of each regime,

$$\begin{aligned} \hat{P}_h^{\text{max},i} &= \frac{s_h^i (T_h^i - T)}{m_h (T_h^i - T)} = \frac{s_h^i}{m_h}, \quad \forall T < T_h^i, \\ \hat{P}_c^{\text{max},i} &= \frac{s_c^i (T - T_c^i)}{m_c (T - T_c^i)} = \frac{s_c^i}{m_c}, \quad \forall T > T_c^i. \end{aligned} \quad (10)$$

The temperature cancels, so the estimate needs no choice of evaluation day. The net-load fit supplies the numerator, which the DSO

observes. The pilot supplies only the denominator m_k . The estimate is therefore only as good as the transferred sensitivity, which Section IV tests.

The two regimes are estimated separately. Their sum equals the total installed ETL capacity only when no device serves both regimes. A reversible HP counts in both $\hat{P}_h^{\text{max}}$ and $\hat{P}_c^{\text{max}}$. For a population of reversible HPs, the total therefore lies between the larger of the two and their sum.

C. Flexibility Quantification

The flexibility an ETL population offers is the energy it can shift from its baseline consumption without breaching indoor thermal comfort [38]. Building-stock insulation and indoor-comfort data are lacking, so an abstract time window Δt_{Th} stands in. Within it, the population must deliver a fixed thermal energy to stay comfortable. A larger Δt_{Th} means a better-insulated building, or wider comfort bands [49]. Over this Δt_{Th} , the aggregated ETLs must deliver the temperature-set energy demand. This is the temperature-driven ETL energy of (8), evaluated over the window Δt_{Th} and drawn at the baseline power $\hat{P}_{\text{ETL}}(T) = \hat{S}\hat{F}(T) \hat{P}^{\text{max}}$,

$$\hat{E}_{\text{ETL}}^{\Delta t_{Th}} = \hat{P}_{\text{ETL}}(T) \Delta t_{Th} = \hat{S}\hat{F}(T) \hat{P}^{\text{max}} \Delta t_{Th}, \quad (11)$$

where $\hat{S}\hat{F}$ and $\hat{P}^{\text{max}}$ are those of the active regime, from (9) and (10). Their product equals the fitted ETL load of (3), so the baseline is the observable temperature-driven load. The schedule that delivers this same energy while deviating most from the baseline is the two-level square wave of Fig. 4: full installed power $\hat{P}^{\text{max}}$ for a fraction α_F of the window and zero for the rest. Its energy $\alpha_F \hat{P}^{\text{max}} \Delta t_{Th}$ must equal (11), which fixes the duty cycle at the simultaneity factor,

$$\alpha_F \hat{P}^{\text{max}} \Delta t_{Th} = \hat{S}\hat{F}(T) \hat{P}^{\text{max}} \Delta t_{Th} \implies \alpha_F = \hat{S}\hat{F}(T). \quad (12)$$

The flexible energy is this schedule's symmetric deviation from the baseline. During the on phase the population draws $\hat{P}^{\text{max}} - \hat{P}_{\text{ETL}}$ above baseline for a time $\alpha_F \Delta t_{Th}$; during the off phase it curtails $\hat{P}_{\text{ETL}}$ for a time $(1 - \alpha_F) \Delta t_{Th}$. The two areas are equal, giving an upward and downward flexible energy

$$E_{\text{flex}} = \hat{S}\hat{F}(T) (1 - \hat{S}\hat{F}(T)) \hat{P}^{\text{max}} \Delta t_{Th}. \quad (13)$$

This use case draws on both fits and one external parameter. Since $\hat{S}\hat{F}(T) \hat{P}^{\text{max}}$ equals the fitted ETL load, (13) equals $\hat{P}_{\text{ETL}}(T) (1 - \hat{S}\hat{F}(T)) \Delta t_{Th}$. The net-load fit sets the load that can be shifted, and the transferred sensitivity only the headroom above it, so the capacity estimate cancels. The window Δt_{Th} , over which the population can be reshaped, is a property of the building stock, not of the fits. The

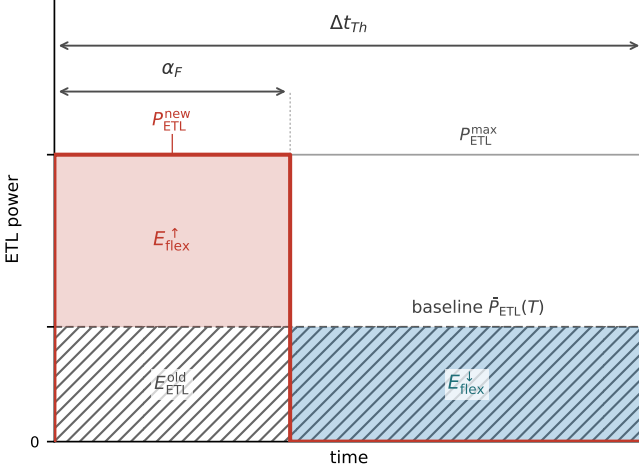


Fig. 4. The flexibility of (13) over one cycle of the square-wave schedule. The aggregate must draw the baseline energy $E_{\text{ETL}}^{\text{old}} = \bar{P}_{\text{ETL}}(T) \Delta t_{Th}$ (hatched) over the window. It can instead draw the same energy as the full-power pulse $P_{\text{ETL}}^{\text{new}}$ of duty cycle $\alpha_F = \hat{\text{SF}}(T)$. The deviation from baseline is the flexible energy, equal upward ($E_{\text{flex}}^{\uparrow}$) and downward ($E_{\text{flex}}^{\downarrow}$).

instantaneous headroom $(1 - \hat{\text{SF}}(T)) \hat{P}^{\text{max}}$ is the special case of the on phase drawn to full power.

Over a year, this paper counts one such window per day of R1 or R3, the regimes in which (9) is defined. The annual flexible energy is then the sum of (13) over these days, each evaluated at its daily mean temperature, and it scales linearly with Δt_{Th} . The days of R2 are left out of the estimate and of its ground truth alike. The estimate therefore covers only temperature-driven flexibility; weather-independent use that could also be shifted, such as hot-water heating, falls outside it.

The bound is an idealisation, not a dispatchable schedule. It takes the square wave as feasible and ignores each dwelling's thermal state and the comfort limits within which its ETLs can shift. It therefore stands as a potential against temperature and window length, not as a firm reserve.

IV. NUMERICAL RESULTS

A. Kloten Data and Evaluation Design

This section uses a single Swiss population, the households metered at the Kloten weather station in calendar 2023. The data combines two sources, the HEAPO dataset [43] and an additional smart-meter dataset [44]. The load profiles have 15-minute resolution, while the ambient temperature is hourly. The only ETL used throughout is the HP and only heating is observed. The station holds 50 households with a submetered HP and 1307 households with neither an HP nor any other electric heating.

The 50 HPs are split at random into two halves of 25, twenty times over. In each split, the first half is the submetered pilot: its aggregated SF is fitted with (7) and supplies the sensitivity m_h . Synthetic aggregates are then drawn from the second half only, so no HP informs both the pilot and a target. Each aggregate combines N_{hp} distinct HPs of that half, with the same homes' other load, and $N - N_{hp}$ distinct HP-free households. The consumer count N takes the values 10, 20, 30, 40 and 50, and N_{hp} the values 5, 10, 15, 20 and 25, with $N_{hp} \leq N$. Every feasible cell holds three aggregates per split, giving 1260 aggregates. All draws are without replacement, so no household recurs within an aggregate.

Two error metrics recur below. The weighted absolute percentage error (WAPE) scores an estimate $\hat{y}_i$ against its ground truth y_i across the aggregates i ,

$$\text{WAPE} = \frac{\sum_i |\hat{y}_i - y_i|}{\sum_i y_i} \times 100\%, \quad (14)$$

so WAPE weights each aggregate by its true magnitude, not equally. A small aggregate, with a large percentage error but a small absolute error, therefore cannot dominate. The metric reports the total error across a set of aggregates, as a DSO would read it. The median ratio, the median of $\hat{y}_i/y_i$, gives the direction of the error, < 1 for an underestimate and > 1 for an overestimate.

B. Fit Quality across Spatial and Temporal Aggregation

Both fits, (2) and (7), are made to each aggregate's measurements, varying the number of consumers and the resolution. The ETL rate is kept fixed at 50%, on a separate set of Kloten aggregates, also drawn without replacement. Spatial aggregation averages out the intermittent, non-thermal behaviour, as more meters are pooled behind one aggregate. Temporal aggregation does the same over time, averaging the load over longer windows removes the occupancy swings and building thermal inertia that do not track temperature one-to-one at finer resolutions. Fig. 5 reports both, as the median R_h^2 of the net-load and SF fits over every consumer-count/averaging-window cell at this fixed ETL rate. Both fits are scored on the points below the net-load threshold T_h , at which the SF fit is anchored.

Each column carries the spatial effect, read from ten consumers at the bottom of the grid to fifty at the top. In the hourly column the net-load fit rises from a median R_h^2 of 0.42 at ten consumers to 0.64 at fifty, and the SF fit rises the same way, from 0.60 to 0.78. The same rise holds in the daily column, where the net-load fit climbs from 0.83 to 0.91 and the SF fit from 0.92 to 0.95.

Each row carries the temporal effect, read from hourly resolution on the left to daily on the right. At ten consumers the net-load fit climbs monotonically from a median R_h^2 of 0.42 at hourly resolution through 0.58 and 0.66 at four and eight hours, to 0.76 at sixteen and 0.83 at daily; the SF fit climbs the same way, from 0.60 through 0.78 and 0.86 to 0.91 and 0.92. The same climb holds in every other row: at fifty consumers, for instance, the net-load fit reaches 0.87 and 0.91 in the two rightmost cells and the SF fit reaches 0.94 and 0.95. The daily mean, used throughout the rest of the paper, therefore sits at the right edge of this curve, with sixteen hours already close behind it.

The two effects compound along the grid's diagonal. The weakest cell, ten consumers at hourly resolution, gives a median R_h^2 of 0.42 for the net-load fit and 0.60 for the SF fit; the strongest, fifty consumers at daily resolution, reaches 0.91 and 0.95. This suggests the two forms of aggregation are, to some extent, substitutable: a coarser averaging window recovers much of what a larger aggregate would, and a larger aggregate recovers much of what a coarser window would.

The heating sensitivity itself scales with the population behind the meter. Fig. 6 plots the fitted s_h against the installed HP capacity across the aggregates, coloured by the number of HPs. The two track closely, at a Pearson correlation of 0.97, along a line through the origin of slope 0.020°C^{-1} . A steeper R1 arm therefore signals more installed capacity, as Section II-A anticipated. This slope exceeds the SF sensitivity of Section IV-C by about 12%, a first sign of the non-ETL temperature response that the net-load sensitivity absorbs.

C. Simultaneity Factor: Invariance and Transfer

Beyond fitting well, the SF fit is stable. Because the SF comes from the submetered HPs alone, the non-HP consumers do not affect

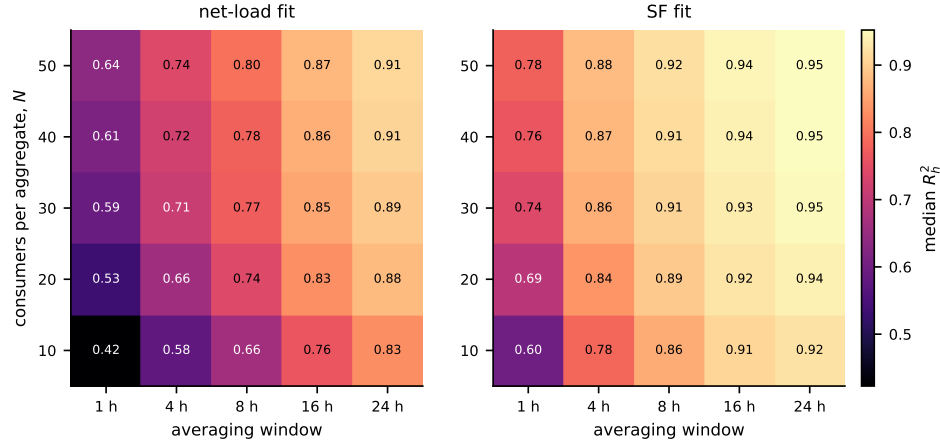


Fig. 5. Median R_h^2 of the net-load fit (left) and the SF fit (right) of the Kloten aggregates against spatial aggregation (consumers per aggregate, N) and temporal aggregation (averaging window), at a fixed 50% ETL rate.

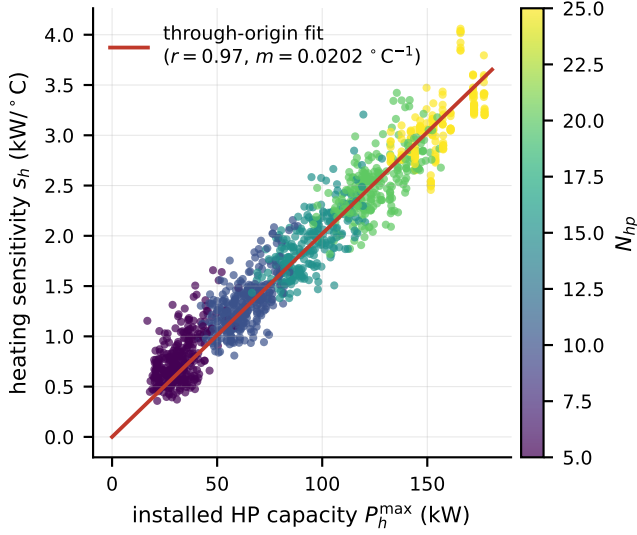


Fig. 6. Fitted heating sensitivity s_h against installed HP capacity across the 1260 Kloten aggregates, coloured by the number of HPs.

it. Table III therefore reports the aggregates of a constant size, 50 consumers, where N_{hp} is the only variable. Neither the maximum heating SF nor the sensitivity m_h changes meaningfully with the number of HPs or their installed capacity: m_h holds near $0.018\text{ }^{\circ}\text{C}^{-1}$ and the maximum SF near 0.48. More HPs behind the meter do not shift these values but tighten them, as the interquartile range of m_h narrows from 0.0047 at five HPs to 0.0013 at twenty-five. This coldest-day value is of the same order as the HP SF of 0.425, drawn from field trials, that national LV network studies use for planning [50].

The pilot carries the SF to the aggregates without submetering. Fig. 7 shows the pilot of one split, whose SF fit, at the pilot's own net-load threshold, gives $\text{SF}_h(T) = 0.021 + 0.019(16.6 - T)$ for $T < 16.6\text{ }^{\circ}\text{C}$, at an R_h^2 of 0.94. Across the twenty splits, the pilot sensitivity m_h has a median of $0.018\text{ }^{\circ}\text{C}^{-1}$ and an interquartile range of 0.017–0.019. This sensitivity is the only parameter that (10) and the flexibility estimate carry to a target.

A direct test sets the pilot against the whole other half, over 120 random splits. Placed at the other half's net-load threshold, the pilot's

TABLE III
SF FITS OF THE KLOTEN AGGREGATES, MEDIAN (IQR)

N_{hp}	R_h^2	$\text{SF}_h^{\max}$	b_h	$m_h\text{ (}^{\circ}\text{C}^{-1}\text{)}$
5	0.92 (0.04)	0.48 (0.11)	0.022 (0.028)	0.0182 (0.0047)
10	0.94 (0.02)	0.47 (0.08)	0.025 (0.016)	0.0177 (0.0036)
15	0.95 (0.02)	0.47 (0.06)	0.027 (0.010)	0.0177 (0.0023)
20	0.95 (0.01)	0.48 (0.05)	0.026 (0.009)	0.0183 (0.0018)
25	0.95 (0.01)	0.48 (0.03)	0.026 (0.007)	0.0180 (0.0013)

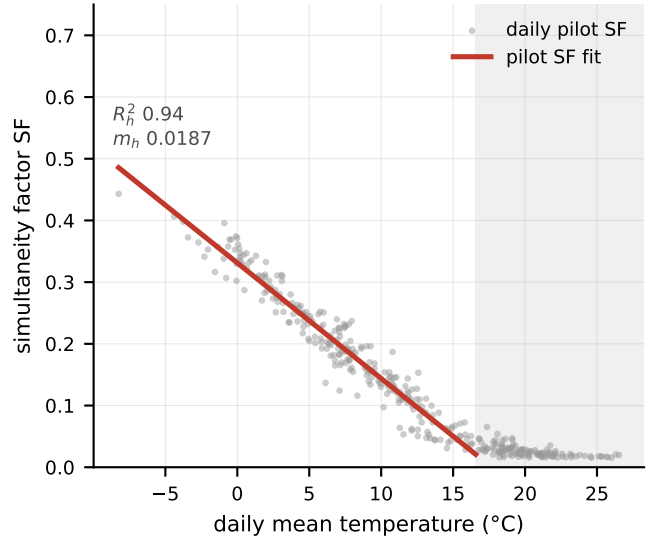


Fig. 7. SF fit to the aggregated pilot half of the Kloten HPs in one random split (grey), at an R_h^2 of 0.94. Its sensitivity m_h is the parameter transferred to the target aggregates.

SF curve reproduces the SF of that half at a median R_h^2 of 0.91, with an interquartile range of 0.85–0.94. The pilot's sensitivity also recovers the capacity of the other half through (10). The median error is +8%, with an interquartile range from 0 to +18%. Section IV-E uses the pilot sensitivity to estimate installed capacity across the aggregates.

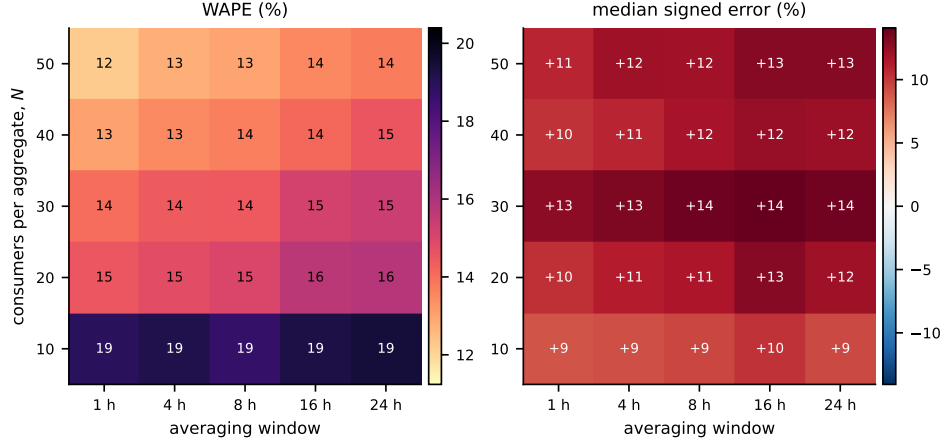


Fig. 8. Capacity error of (10) on the Kloten aggregates against spatial aggregation (consumers per aggregate, N) and temporal aggregation (averaging window), at a fixed 50% ETL rate. Left: WAPE; right: median signed error.

D. Energy Consumption Estimation

Equation (8) needs only the net-load fit, so the energy is the one estimate that requires no pilot. On the days below the threshold, R1 is compared against the submetered HP energy of each aggregate. Across the 1260 aggregates, the estimate reaches a WAPE of 9% at a median ratio of 0.97. About 7% of the annual HP energy falls in R2, most plausibly domestic hot water [11], which (8) omits by construction, so the annual ratio falls to 0.90. Integrating the daily-fitted arm over the hourly temperature, rather than the daily mean, recovers part of this, lifting the annual ratio to 0.94, since a day that crosses the threshold holds more heating-degree-hours than its mean implies. The same R1 energy is the baseline of the flexibility in Section IV-F.

E. Installed Capacity Estimation

Equation (10) estimates the installed HP capacity of each aggregate as the ratio of two sensitivities. The numerator s_h is read from the aggregate's own net-load fit. The denominator m_h is the sensitivity of the pilot of the same split (Section IV-C). The target aggregate is therefore never submetered.

With the transferred sensitivity, (10) recovers the installed capacity of the 1260 aggregates at a WAPE of 19%, with a median overestimate of 16%. In absolute terms, the mean error is 17 kW, the capacity of about three HPs of the mean 6.3 kW. The overestimate is consistent with the non-ETL temperature response that the net-load sensitivity absorbs, the neglected term of (1) discussed in Section VI. The error shrinks as HPs are added: the WAPE falls from 41% at five HPs to 15% at twenty-five, and the median overestimate from 35% to 12%. This is possibly because the HP-free households, whose temperature response inflates the heating arm, weigh less against more HPs.

Fig. 8 repeats the estimate over the aggregation grid of Section IV-B, at a fixed 50% ETL rate. In each cell, both the target's net-load fit and the pilot's SF fit are made at the cell's resolution. The averaging window barely matters: along each row, the WAPE varies by at most 1.5 points, as both sensitivities of (10) come from fits at the same resolution. The consumer count matters more. The WAPE falls from 19% at ten consumers to between 12% and 14% at fifty, while the median overestimate stays between 9% and 14%. Five HPs among ten consumers thus give a WAPE of 19%, against 41% for five HPs across all consumer counts, which points to the share of HP-free households rather than the HP count. Unlike the fit

quality of Fig. 5, the capacity estimate needs no resolution finer than the daily mean.

F. Flexibility Potential

The capacity estimate enters (13) only in appearance. Since $\hat{S}F(T) \hat{P}^{\max}$ equals the fitted ETL load, the estimate reduces to $\hat{P}_{\text{ETL}}(T) (1 - \hat{S}F(T)) \Delta t_{Th}$: the temperature-driven load of the net-load fit, times the headroom left by the transferred SF. The actual flexibility is computed at hourly resolution: each hour's submetered SF gives $SF(1 - SF) P_h^{\max}$ with the true capacity, averaged over the day. Both are summed over the days of R1, below the aggregate's net-load threshold, since the Kloten aggregates have no cooling regime. The estimate and its ground truth share no input other than the HP load contained in the net load, so their agreement does not follow by construction.

Fig. 9 gives the distribution of the estimated-to-actual ratio of annual flexible energy across the 1260 aggregates. The ratio has a median of 1.04 and an interquartile range of [0.97, 1.17], at a WAPE of 10%. It splits into two factors. The R1 energy of Section IV-D contributes a median ratio of 0.97, and the headroom a median ratio of 1.07. About three points of the headroom factor come from the transferred SF running slightly below the measured one, and about four from the daily resolution of the estimate, since the SF swings within the day. The flexibility and energy ratios correlate at 0.96. The flexibility estimate therefore inherits its accuracy mainly from the net-load fit, while the transferred sensitivity shifts the headroom by a few per cent. Since Δt_{Th} cancels in the ratio, neither factor depends on the window.

Computed from the daily mean SF instead, as the estimate is, the actual flexibility would be 3% higher and the median ratio 1.01, since $SF(1 - SF)$ is concave. The days of R2, left out by construction, would add a weather-independent floor, most plausibly hot-water heating, worth a median 9% of the actual flexible energy over the whole year. On the coldest day, the estimate recovers the flexible energy at a WAPE of 21% and an R^2 of 0.73 against the actual values across aggregates.

Fig. 10 shows the daily actual and estimated flexible energy over the year for the held-out half of one split: 25 HPs with $P_h^{\max} = 144$ kW, drawn for an illustrative window $\Delta t_{Th} = 4$ h. None of these HPs informs the pilot, so the example is out of sample. The potential follows the season and is largest in winter, when the thermal demand that can be shifted is largest. The estimate tracks this profile

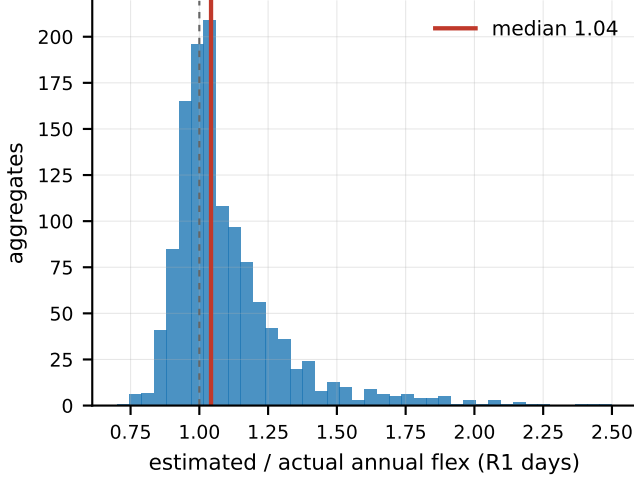


Fig. 9. Distribution across the 1260 Kloten aggregates of the estimated-to-actual ratio of annual flexible energy on the R1 days, with the median marked. The estimate uses the net-load fit and the transferred sensitivity; the actual uses each aggregate’s own hourly submetered SF and true capacity.

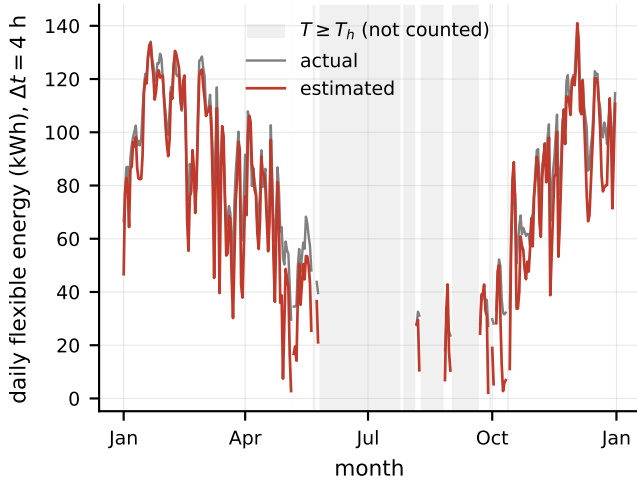


Fig. 10. Daily flexible energy of a held-out half of the Kloten HPs (25 HPs, $P_h^{\max} = 144$ kW) over the year, for an illustrative window $\Delta t_{Th} = 4$ h; since E_{flex} is linear in Δt_{Th} , the window sets only the vertical scale. Estimated from the pilot sensitivity of the same split versus computed from the hourly submetered SF; the shaded days of R2 are not counted.

day to day. A third of the days lie above the threshold, mostly from late spring to early autumn, and fall outside the count. On them, the measured flexibility keeps a floor of about 19 kWh per day, near 14% of its winter maximum, which the estimate omits by construction. On the coldest days the $SF(1 - SF)$ product nears its 0.25 ceiling, a quarter of $P_h^{\max} \Delta t_{Th}$, in both the actual and the estimate. Over the R1 days of the year, the flexible energy rises by 5.2 MWh per hour of window for the actual and 4.7 MWh for the estimate.

V. REAL AGGREGATES

Section IV tested the method within one population, on aggregates built from its households. This section applies it to real aggregates that no pilot has seen, each summed from its own field meters. Swiss and German aggregates test the heating regime with the Kloten pilot. Submetered data serve only to validate the results, never as input.

TABLE IV
TRANSFER TO REAL AGGREGATES OUTSIDE KLOTEN

aggregate	N_{hp}	$P_h^{\max}$ (kW)	$\hat{P}_h^{\max}$ (kW)	cap. (%)	flex
Hg, CH	21	109	112	+2	0.96
MqO, CH	6	51	48	-6	0.93
Hamelin, DE	37	239	217	-9	0.93

Three real aggregates lie outside the Kloten pilot. Two are the HEAPO households of two further Swiss weather stations, Hg and MqO, with 21 and 6 submetered HPs. The third comes from the WPuQ measurement campaign in Hamelin [42]: its net load is the sum of 37 individually metered German single-family houses, each with a submetered HP. All three are summed rather than metered at a transformer, so they carry no network losses. They also contain no consumers without an HP, which makes their ETL rate 100%. The pilot is now the full Kloten pool of 50 HPs. Its SF fit, at the pool’s net-load threshold, gives $SF_h(T) = 0.029 + 0.018(16.6 - T)$, and only its sensitivity $m_h = 0.018^\circ\text{C}^{-1}$ transfers.

Table IV reports the three estimates. The capacity errors are +2% at Hg, -6% at MqO and -9% at Hamelin. MqO holds only six HPs, below the ten from which Section IV-E finds the estimates reliable, so its result is indicative. The flexibility ratio on the R1 days lies between 0.93 and 0.96, as in Section IV, and follows the R1 energy ratios of 0.89 to 0.92.

The thresholds show why only the sensitivity transfers. The net-load thresholds are 16.7°C at Hg, 14.0°C at MqO and 13.9°C at Hamelin, against 16.6°C for the Kloten pilot. The threshold temperature varies with building fabric, internal gains and setpoint preference [51]. Transferring the whole Kloten curve, intercept and threshold included, gives capacity errors of -3%, -20% and -26%. The error of the whole curve therefore grows with the distance between the target’s threshold and the pilot’s, whereas the sensitivity alone stays within 10%.

VI. DISCUSSION

A. Accuracy

The capacity estimate carries two error sources. The first is the non-ETL temperature response that (1) neglects, which biases the net-load numerator upwards. A zero-HP control makes this concrete. Aggregates of 10 to 50 HP-free Kloten households still show a small heating arm, from which (10) infers a median 0.28 kW of spurious capacity per household. This floor is about 4% of the mean installed capacity of one HP, 6.3 kW. It therefore adds about 45% to the estimate at a 10% ETL rate, but only about 4% once every consumer has an HP.

The second source is the transferred sensitivity. Within the Kloten population, the pilot’s sensitivity matches the aggregates’ own at a median ratio of 1.00, with an interquartile range of $\pm 10\%$, so the transfer adds spread rather than bias. On the real aggregates outside the pilot, the estimates lie within 10% of the measured capacity.

B. Generalisation

The SF generalises in a specific and testable way. Within a population, its sensitivity is empirically stable against the number of HPs and their installed capacity over the tested range, so a curve from a submetered pilot serves that population’s aggregates without submetering. Across populations, only the sensitivity transfers. The threshold temperature follows the local building stock and climate, and the net-load fit supplies it from data the DSO already meters.

The intercept describes weather-independent ETL use, which lies outside the temperature-driven quantities estimated here. Transferring the whole curve instead of the sensitivity alone raises the capacity error to 20% at MqO and 26% at Hamelin.

The sensitivity itself carries between populations of the same climate class. The Kloten pilot's $0.018\text{ }^{\circ}\text{C}^{-1}$ lies within 12% of the own sensitivities of Hg, MqO and Hamelin. Across climate classes, however, the sensitivity varies by up to a factor of four, and within one class it can still differ between HP technologies (Table II). Generalisation therefore reduces to transferring a single parameter from a pilot of the same climate class and HP technology. This mirrors the inverse problem solved for PV installed capacity against irradiance [23], here posed against ambient temperature.

C. Data Requirements

The two fits ask for different data. The net-load fit needs only the aggregated net load and ambient temperature, both already held by the DSO, and runs on any aggregate. The SF fit needs submetered ETL load and installed capacity, which the DSO rarely holds. A single submetered pilot of the same population calibrates it, and its sensitivity then transfers, so no aggregate under study is submetered. A pilot of a few tens of HPs appears to suffice: a pilot of 25 Kloten HPs reproduces the SF shape of the other 25 at a median R_h^2 of 0.91, and recovers their capacity at a median error of +8% (Section IV-C). The pilot should, however, submeter every electric heating source of its homes, including backup resistance heat.

D. Limitations

Several limits temper the results. The within-population tests rest on the 50 submetered Kloten HPs. The synthetic aggregates are drawn without replacement and never share an HP with their pilot, but aggregates of the same split draw on the same 25 HPs and therefore overlap. The cross-population tests rest on three real aggregates in the heating regime, one of them with only six HPs. The cooling regime remains untested. The model neglects the non-ETL temperature response throughout, which the bias at a low HP count exposes.

A further confounder is deliberate control. Swiss and German domestic HPs commonly run under interruptible tariffs with defined blocking periods, and since January 2024 §14a EnWG lets German DSOs throttle HPs above 4.2 kW [52]. Such control truncates the aggregated ETL peak. It may bias the net-load numerator and the capacity estimate downwards, and imprints regime-specific structure that a single-sensitivity transfer need not capture. On the real aggregates, however, the estimates lie within 10% of the measured capacity, so such control is not a visible source of error there. The WPuQ campaign predates §14a, but blocking-period tariffs do not. Underlying all of these is the broader shortage of submetered ETL data, which a wider pilot would relieve.

E. Concluding Remarks

The method trades a modest, characterised error for a large gain in deployability. Where a submetered pilot of the population exists, the installed capacity of a real aggregate of tens of HPs follows to within about ten per cent, and its energy and flexibility from the same fits. Where it does not, the net-load fit alone still bounds the temperature-driven energy. The estimates are least reliable on aggregates with a handful of HPs and most reliable once tens aggregate, which is the regime a substation presents.

VII. CONCLUSION

This paper presented a method to quantify an aggregated ETL population from data available to DSOs. Two temperature-response fits were made, an observable net-load fit and an SF fit calibrated on a submetered pilot. Three planning quantities were derived from them: installed capacity, temperature-driven energy and a flexible-energy potential. The installed capacity follows per regime as the ratio of the net-load sensitivity to the SF sensitivity, the only parameter that transfers from the pilot. The SF sensitivity proved empirically stable against the number of HPs and their installed capacity within a population. Across synthetic Kloten aggregates of 5 to 25 HPs, the method recovered the installed capacity to a WAPE of 19%, falling to between 12% and 14% for aggregates of fifty consumers at a 50% ETL rate. On three real Swiss and German aggregates outside the pilot, it came within 10% of the measured capacity, whereas transferring the whole SF curve erred by up to 26%.

Regarding future work, four directions follow. The submetered pools were small, so a wider and more diverse set of metered HPs, with every electric heating source submetered, would test the transfer on more populations and in the cooling regime. The net-load sensitivity also absorbs the temperature response of the non-ETL load, which inflates the capacity estimate most where consumers without ETLs dominate. Future work will compensate the estimates for this response, for instance by subtracting a per-consumer non-ETL sensitivity scaled by the consumer count the DSO already holds. It will also evaluate data-driven methods for HP and ETL quantification against the change-point estimator, to see where a learned model earns its data cost and where the two-parameter fit suffices. Finally, it will address the dead-band limit by modelling the non-temperature-driven ETL consumption, chiefly domestic hot water and baseline operation. This consumption is what the heating arm omits, and it sets the gap between the temperature-driven and the annual energy.

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